



Thiagarajar College of Engineering, Madurai-625 015.
(A Govt. Aided Autonomous Institution Affiliated to Anna University)

Department of Mathematics

Assignment – I

Program / Branch: B.E./EEE

Subject Code:22EE410

Subject Name: Probability and Statistics

Slot: A & B

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Answer all questions

Apply Level Questions

1. The number of pumps in use at both a six-pump station and a four-pump station will be determined. Give the possible values for each of the following random variables: **a.** T = the total number of pumps in use **b.** X = the difference between the numbers in use at stations 1 and 2 **c.** U = the maximum number of pumps in use at either station **d.** Z = the number of stations having exactly two pumps in use .

2. Airlines sometimes overbook flights. Suppose that for a plane with 50 seats, 55 passengers have tickets. Define the random variable Y as the number of ticketed passengers who actually show up for the flight. The probability mass function of Y appears in the accompanying table.

y	45	46	47	48	49	50	51	52	53	54	55
$p(y)$	0.05	0.1	0.12	0.14	0.25	0.17	0.06	0.05	0.03	0.02	0.01

a. What is the probability that the flight will accommodate all ticketed passengers who show up? **b.** What is the probability that not all ticketed passengers who show up can be accommodated? **c.** If you are the first person on the standby list (which means you will be the first one to get on the plane if there are any seats available after all ticketed passengers have been accommodated), what is the probability that you will be able to take the flight? What is this probability if you are the third person on the standby list?

3. Some parts of California are particularly earthquake- prone. Suppose that in one metropolitan area, 25% of all homeowners are insured against earthquake damage. Four homeowners are to be selected at random; let X denote the number among the four who have earthquake insurance. **a.** Find the probability distribution of X **b.** Draw the corresponding probability histogram. **c.** What is the most likely value for X ? **d.** What is the probability that at least two of the four selected have earthquake insurance?

4. The article “Should You Report That Fender Bender?” (Consumer Reports, Sept. 2013: 15) reported that 7 in 10 auto accidents involve a single vehicle (the article recommended always reporting to the insurance company an accident involving multiple vehicles). Suppose 15 accidents are randomly selected. **a.** What is the probability that at most 4 involve a single vehicle? **b.** What is the probability that exactly 4 involve a single vehicle? **c.** What is the probability that exactly 6 involve multiple vehicles? **d.** What is the probability that between 2 and 4, inclusive, involve a single vehicle? **e.** What is the probability that at least 2 involve a single vehicle? **f.** What is the probability that exactly 4 involve a single vehicle and the other 11 involve multiple vehicles?

5. An article in the Los Angeles Times (Dec. 3, 1993) reports that 1 in 200 people carry the defective gene that causes inherited colon cancer. In a sample of 1000 individuals, what is the

approximate distribution of the number who carry this gene? Use this distribution to calculate the approximate probability that **a.** Between 5 and 8 (inclusive) carry the gene. **b.** At least 8 carry the gene.

6. Mopeds (small motorcycles with an engine capacity below 50 cm³) are very popular in Europe because of their mobility, ease of operation, and low cost. The article “Procedure to Verify the Maximum Speed of Automatic Transmission Mopeds in Periodic Motor Vehicle Inspections” (J. of Automobile Engr., 2008: 1615–1623) described a rolling bench test for determining maximum vehicle speed. A normal distribution with mean value 46.8 km/h and standard deviation 1.75 km/h is postulated. Consider randomly selecting a single such moped. **a.** What is the probability that maximum speed is at most 50 km/h? **b.** What is the probability that maximum speed is at least 48 km/h? **c.** What is the probability that maximum speed differs from the mean value by at most 1.5 standard deviations?

7. A large but sparsely populated county has two small hospitals, one at the south end of the county and the other at the north end. The south hospital’s emergency room has four beds, whereas the north hospital’s emergency room has only three beds. Let X denote the number of south beds occupied at a particular time on a given day, and let Y denote the number of north beds occupied at the same time on the same day. Suppose that these two rv’s are independent; that the pmf of X puts probability masses .1, .2, .3, .2, and .2 on the x values 0, 1, 2, 3, and 4, respectively; and that the pmf of Y distributes probabilities .1, .3, .4, and .2 on the y values 0, 1, 2, and 3, respectively. **a.** Display the joint pmf of X and Y in a joint probability table. **b.** Compute $P(X \leq 1 \text{ and } Y \leq 1)$ by adding probabilities from the joint pmf, and verify that this equals the product of $P(X \leq 1)$ and $P(Y \leq 1)$. **c.** Express the event that the total number of beds occupied at the two hospitals combined is at most 1 in terms of X and Y , and then calculate this probability. **d.** What is the probability that at least one of the two hospitals has no beds occupied?

8. Two components of a minicomputer have the following joint pdf for their useful lifetimes X and Y :

$$f(x, y) = \begin{cases} xe^{-x(x+y)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases} . \text{ a. What is the probability that the lifetime } X \text{ of the first}$$

component exceeds 3? **b.** What are the marginal pdf’s of X and Y ? Are the two lifetimes independent? Explain. **c.** Given that the lifetime of $Y=2$, compute the probability that the lifetime of $X > 2$.

9. Show that if X and Y are independent rv’s, then $E(XY) = E(X) \cdot E(Y)$.

10. Reconsider the minicomputer component lifetimes X and Y as described in Exercise 9. Determine $E(XY)$. What can be said about $\text{Cov}(X, Y)$ and the correlation?

11. Many older homes have electrical systems that use fuses rather than circuit breakers. A manufacturer of 40-amp fuses wants to make sure that the mean amperage at which its fuses burn out is in fact 40. If the mean amperage is lower than 40, customers will complain because the fuses require replacement too often. If the mean amperage is higher than 40, the

manufacturer might be liable for damage to an electrical system due to fuse malfunction. To verify the amperage of the fuses, a sample of fuses is to be selected and inspected. If a hypothesis test were to be performed on the resulting data, what null and alternative hypotheses would be of interest to the manufacturer? Describe type I and type II errors in the context of this problem situation.

12. The desired percentage of SiO₂ in a certain type of aluminous cement is 5.5. To test whether the true average percentage is 5.5 for a particular production facility, 16 independently obtained samples are analyzed. Suppose that the percentage of SiO₂ in a sample is normally distributed with $\sigma = .3$ and that $\bar{x} = 5.25$. **a.** Does this indicate conclusively that the true average percentage differs from 5.5?

b. If the true average percentage is $\mu = 5.6$ and a level $\alpha = .01$ test based on $n = 16$ is used, what is the probability of detecting this departure from H_0 ? **c.** What value of n is required to satisfy $\alpha = .01$ and $\beta(5.6) = .01$?

13. Let μ denote the true average reaction time to a certain stimulus. For a z test of $H_0: \mu = 5$ versus $H_a: \mu > 5$, determine the P-value for each of the following values of the z test statistic.

a. 1.42 **b.** .90 **c.** 1.96 **d.** 2.48 **e.** -.11

14. The following observations are on stopping distance (ft) of a particular truck at 20 mph under specified experimental conditions ("Experimental Measurement of the Stopping Performance of a Tractor-Semitrailer from Multiple Speeds," NHTSA, DOT HS 811 488, June 2011): 32.1 30.6 31.4 30.4 31.0 31.9. The cited report states that under these conditions, the maximum allowable stopping distance is 30. A normal probability plot validates the assumption that stopping distance is normally distributed. **a.** Does the data suggest that true average stopping distance exceeds this maximum value? Test the appropriate hypotheses using $\alpha = .01$. **b.** Determine the probability of a type II error when $\alpha = 0.01$, $\sigma = .65$, and the actual value of μ is 31. Repeat this for $\mu = 32$. **c.** Repeat (b) using $\sigma = 0.8$ and compare to the results of (b). **d.** What sample size would be necessary to have $\alpha = 0.01$ and $\beta = 0.1$ when $\mu = 31$ and $\sigma = 0.65$?

15. The true average diameter of ball bearings of a certain type is supposed to be .5 in. A one-sample t test will be carried out to see whether this is the case. What conclusion is appropriate in each of the following situations? **a.** $n = 13$, $t = 1.6$, $\alpha = 0.5$ **b.** $n = 13$, $t = -1.6$, $\alpha = 0.5$
c. $n = 25$, $t = -2.6$, $\alpha = 0.5$ **d.** $n = 25$, $t = -3.9$